DIGITAL IMAGE SYNTHESIS

Practice 5

Texture2D - Image

Practice 5-1

Changing the colour of a 2D texture applied to a 3D model

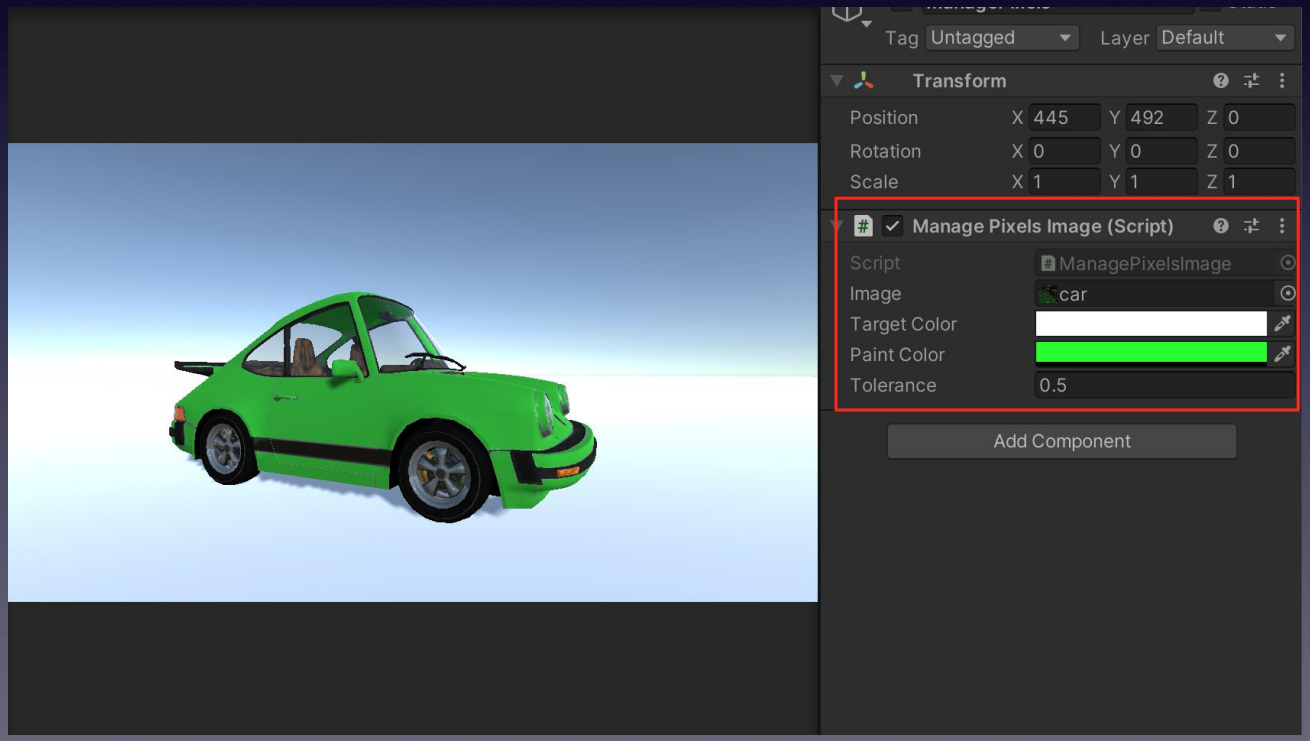
Practice 5-1

- To create a script "ManagePixelsImage" to able change a specific color of a texture2D.

- To achieve this, you must use the "GetPixel " and “SetPixel”method, this allows you to access and modify the color information of each of the pixels in the image.

Practice 5-1

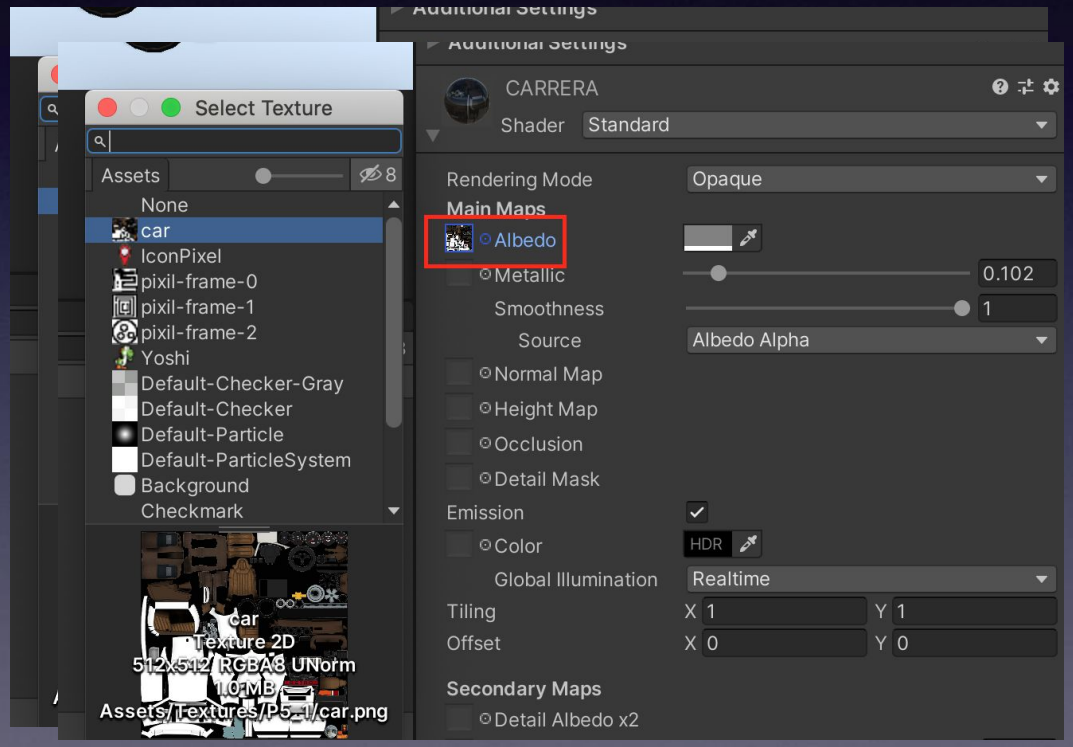
Changing the colour of a 2D texture applied to a 3D model



Practice 5-1

Image - Texture

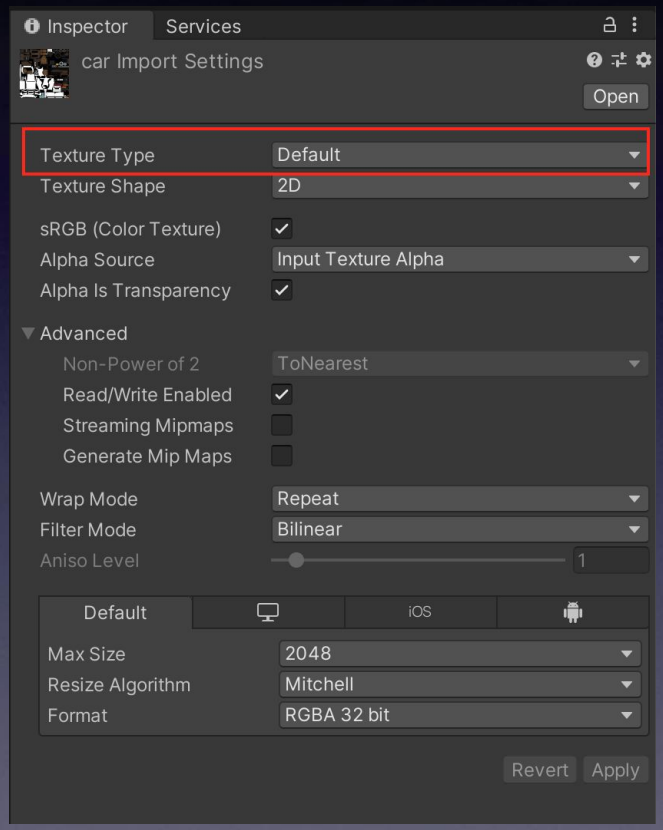
Textures are **bitmap images**. A Material can contain references to textures, so that the **Material's Shader** can use the textures while calculating the surface color of a GameObject. In addition to basic Color (Albedo) of a GameObject's surface, Textures can represent many other aspects of a Material's surface such as its reflectivity or roughness.



Practice 5-1

Image - Texture

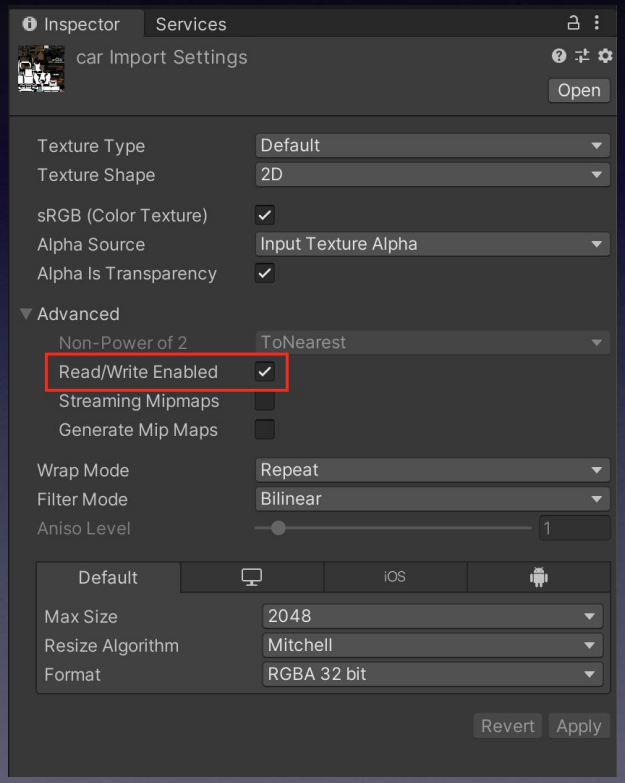
In Unity an image is treated as a texture, there are several types of texture but to apply it to a 3D model we use the "**Default**" type.



Practice 5-1

Image - Texture

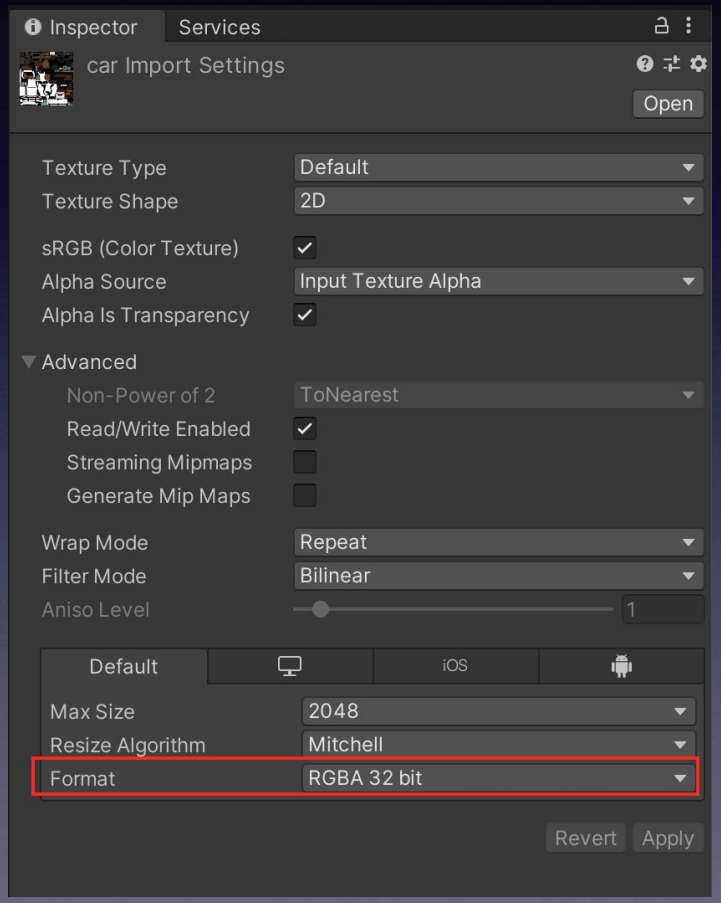
If you want to access and modify the information of an image, the texture attribute **"Read/Write Enabled"** must be **activated**.



Practice 5-1

Image - Texture

And in order to match the information of each pixel to the **RGBA** system, the RGBA format must be applied.



Practice 5-1

Texture2D

To get the **width** and **height** of the texture in pixels

```
public Texture2D image;

private void Awake()
{
    _width = image.width;
    _height = image.height;
}
```

Practice 5-1

Texture2D

When changes are made to an image, it is necessary to call the "**Apply**()" method for them to be applied.

```
image.SetPixel(0, 0, new Color(255,255,0,0));

image.Apply();
```

Practice 5-1

Interesting additional information:

- Even if the image is a matrix of pixels, you can store all the information of each pixel in an array, for example:

```
private Color[] colourMap;

private void Awake()
{
    _width = image.width;
    _height = image.height;

    colourMap = new Color[_width * _height];

}
```

Practice 5-1

Interesting additional information:

- A **tolerance** for color should be specified as e.g. 255.0.0 and 254.0.0 both are considered red. tolerance = 0.5 can be an acceptable threshold.

- You can use the method Vector4.Distance, it calculate the distance between two vector colors.

Practice 5-1

Interesting additional information:

- When you modify an image that has the "read/write enabled" attribute activated, you modify the information of the image loaded in RAM, until you close the project the image will not return to its original state.
- To restore the image to its original state when you stop the execution of the scene, you can use the "OnDisable()" method.

Practice 5-1

Interesting additional information:

- You must use the "**Practice_5.unitypackage**". You will find it in the content of practice 5 in Poliformat.

- In this unitypackage you will find the scene "**P5_1**" you must use it to make this practice.

Practice 5-2

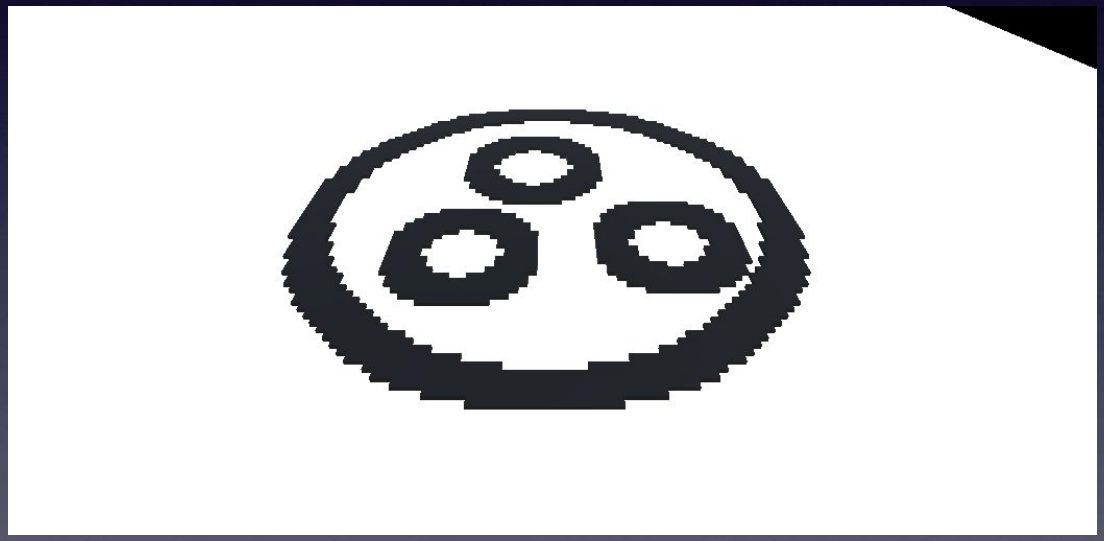
Generate Level From Image

Practice 5-2

Generate Level From Image

Practice 5-2

Generate Level From Image



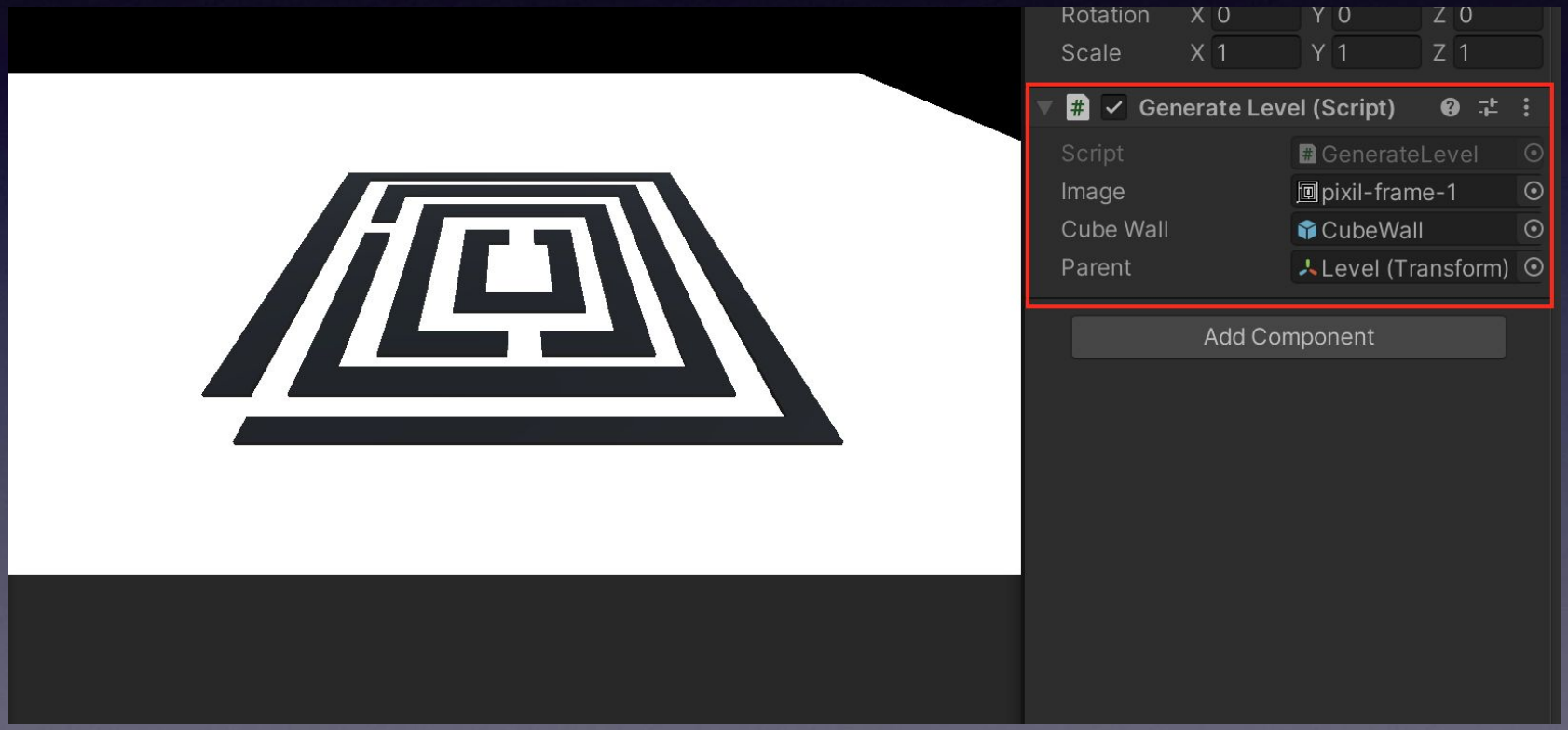
Practice 5-2

- To create a script "**GenerateLevel**" to be able to generate a level assembled with cubes based on the information of an image.

- The relation will be **1 cube = 1 pixel**.

Practice 5-2

Generate Level From Image



Practice 5-2

Interesting additional information:

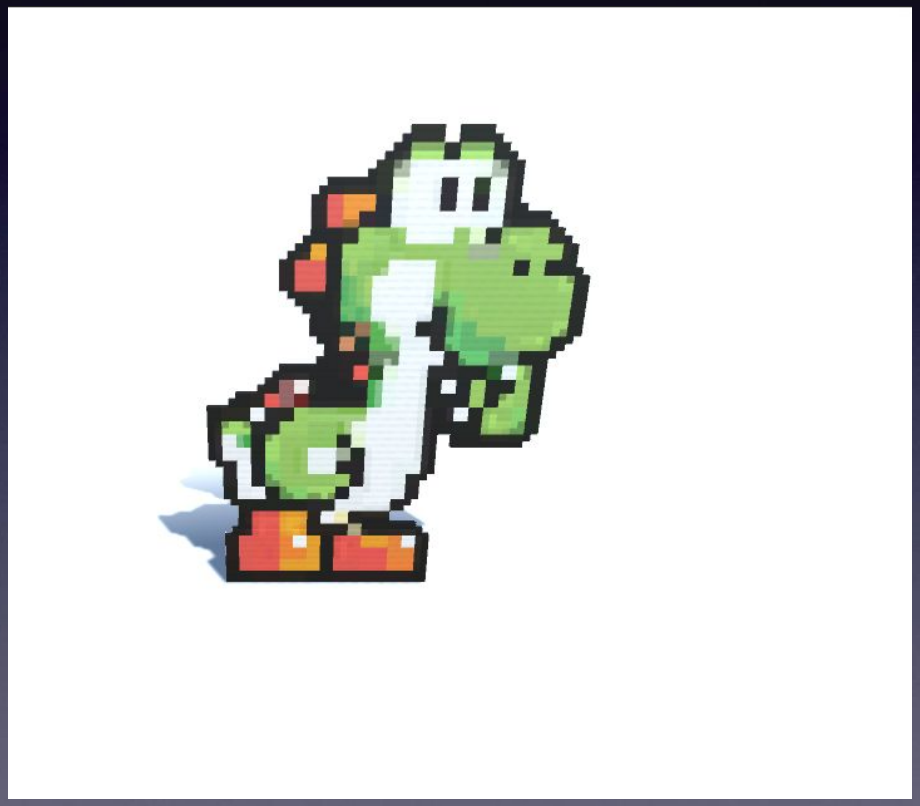
- You must use the "**Practice_5.unitypackage**". You will find it in the content of practice 5 in Poliformat.

- In this unitypackage you will find the scene “**P5_2**” you must use it to make this practice.

- If you want to create your own images you can use: https://www.pixilart.com/draw?ref=home-page

Practice 5-3

Pixel Voxel

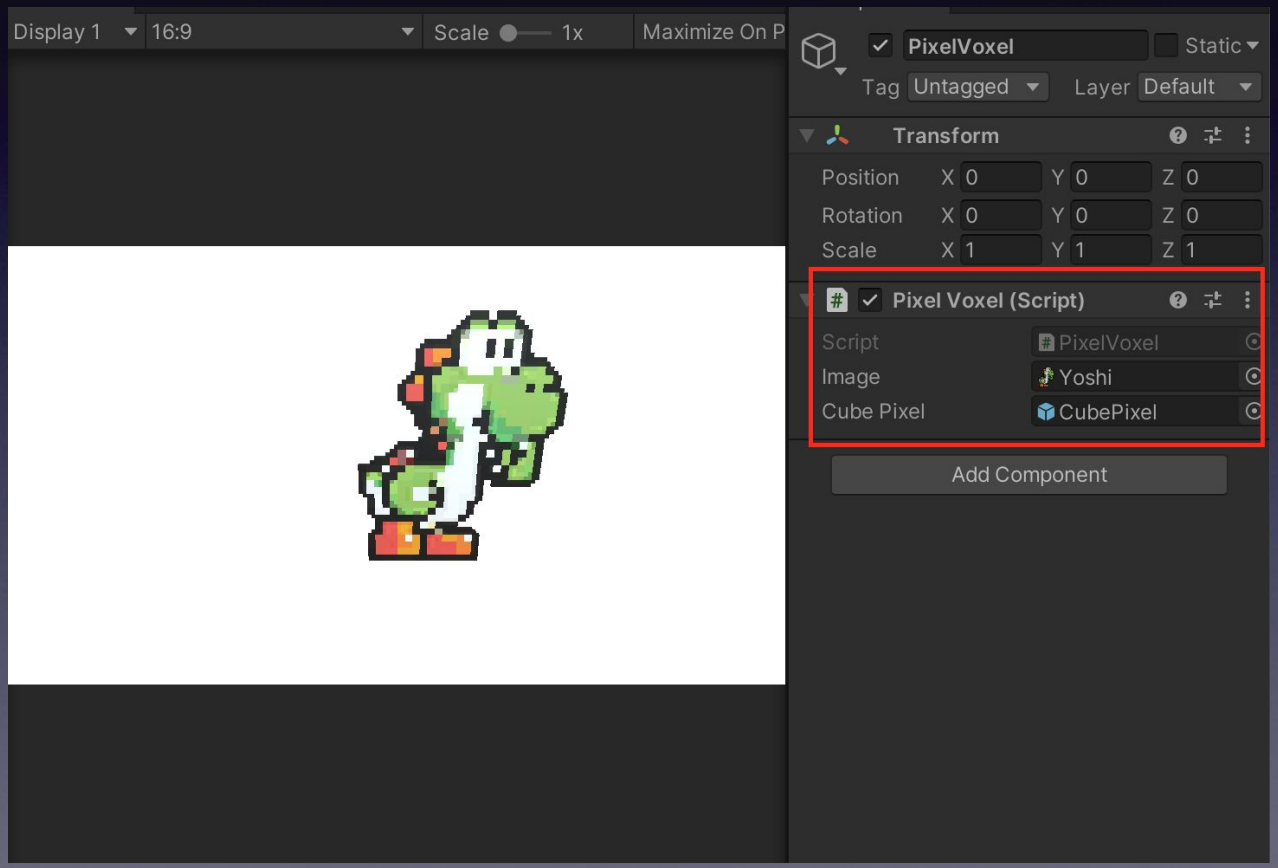


Practice 5-3

- Create a "PixelVoxel" script to represent a 2D image with 3D cubes.
- The relation will be **1 cube = 1 pixel**.

Practice 5-3

Pixel Voxel

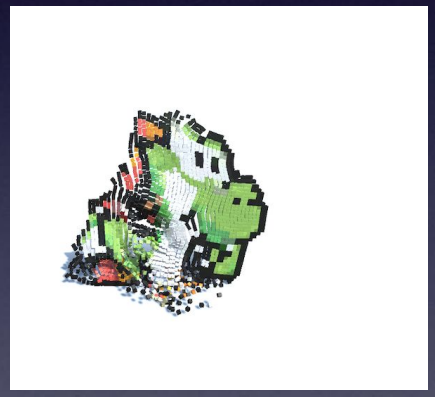


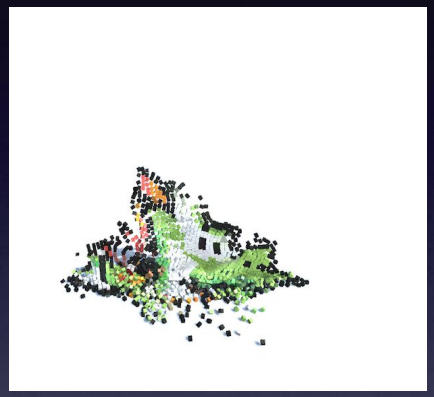
Practice 5-3

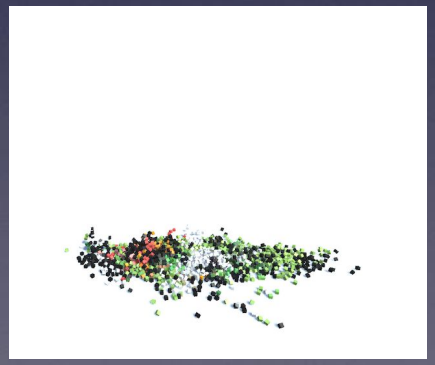
- When you press the space key, the 3D representation of the image must be destroyed using the "Rigidbody" component propertie “**Use Gravity**”

Practice 5-3

Pixel Voxel



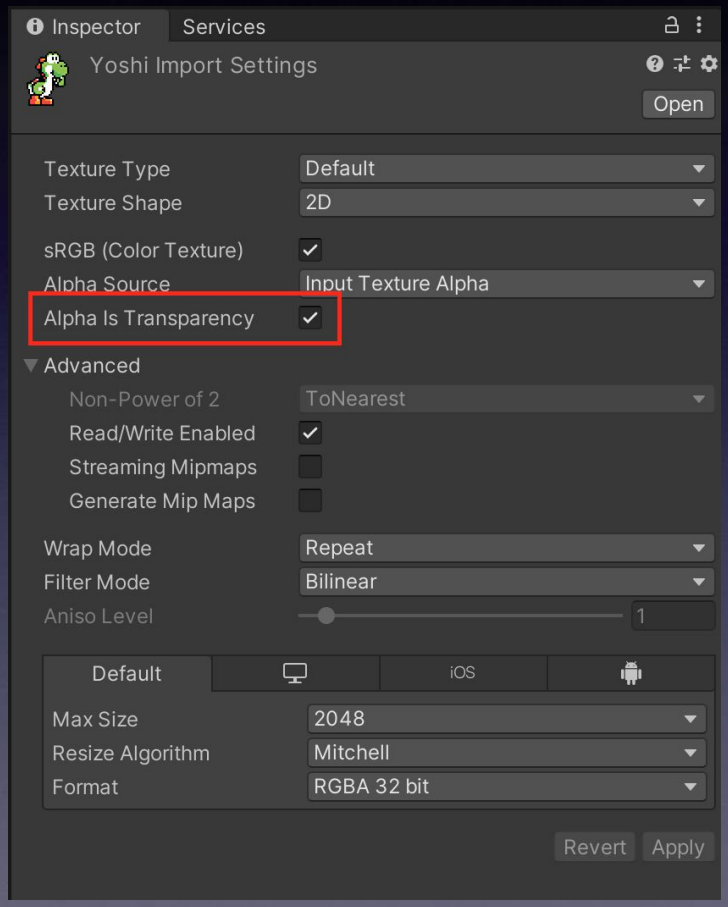




Practice 5-3

Interesting additional information:

- If you want to display the texture's alpha channel as transparent, you must activate the property in the texture “Alpha Is Transparency”



Practice 5-3

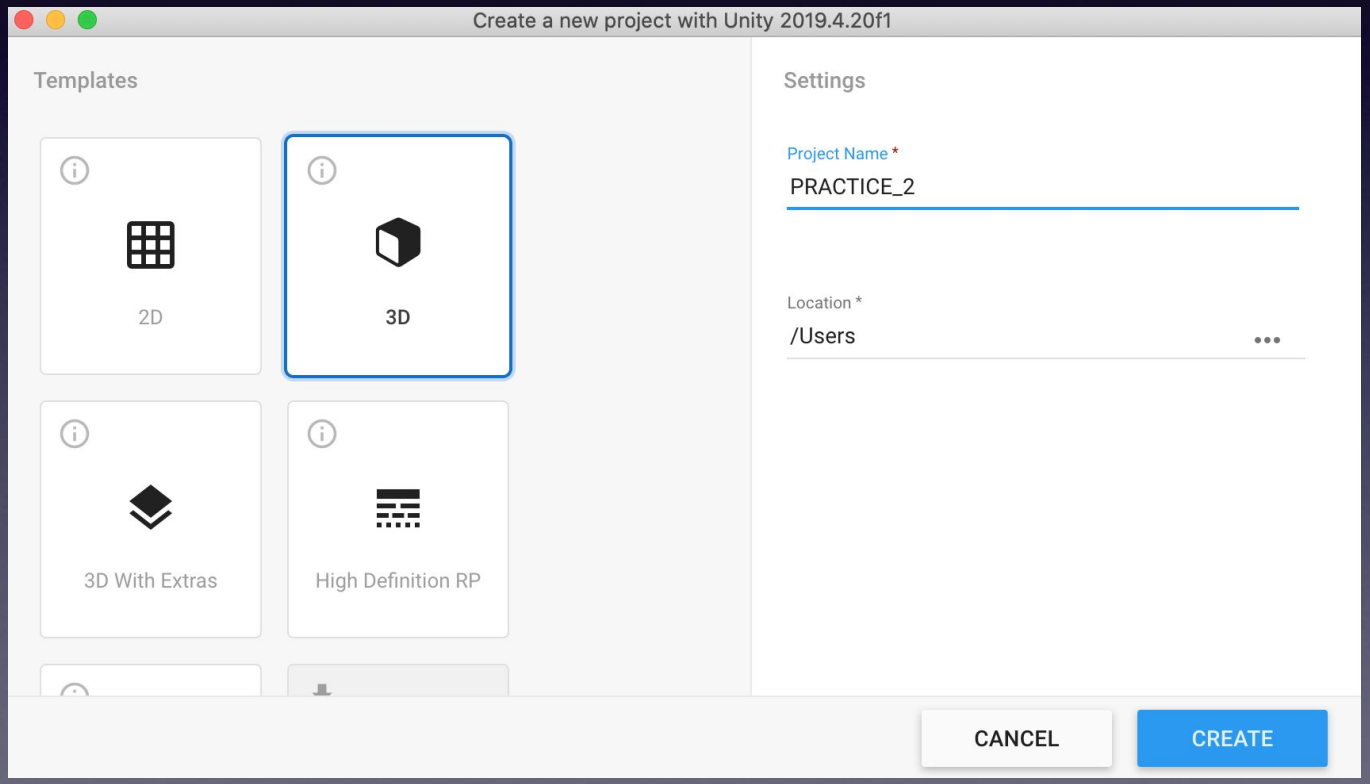
Interesting additional information:

- You must use the "**Practice_5.unitypackage**". You will find it in the content of practice 5 in Poliformat.

- In this unitypackage you will find the scene "**P5_3**" you must use it to make this practice.

Practice 5

Create 3D Project



Practice 5

Send your work

For the assignment of each of the practices, it is necessary to create a private repository in GitLab and give us access to it (user **jorge_orta_lopez**).

The role must be **Reporter** (as Guest I will not be able to download the repository).

And in the **Poliformat task** send me the link to the repository.

Practice 5

Send your work

For further information:

In GitLab's own documentation you can find details on how Git works: here.

You can use Git from the command line or via graphical clients such as GitKraken, Sourcetree, Fork or TortoiseGit.

Personally, I use Sourcetree. You can link it very easily to your GitLab account by following these instructions.

Practice 5

Send your work

For further information:

An important file in the repository is known as **.gitignore**. It defines the files and directories that will be ignored by Git when you upload your changes. In the course, we will see that each Unity project generates several directories that, as a general rule, should never be uploaded to the repository, but should be generated locally on each machine (Library, Temp, Obj, etc.).

It is advisable to take inspiration from the **.gitignore** file available here.